

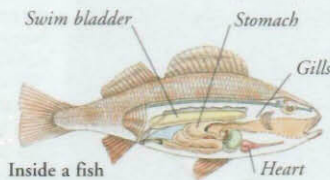
FISH



THE FIRST FISH appeared in the seas 470 million years ago. Today, more than 20,000 species have been described, ranging from the great whale shark to the pygmy goby. Fish live in freshwater streams, rivers, and lakes, and in saltwater seas and oceans. A few, including eels and salmon, migrate from salt to fresh water. Some fish are fierce predators, and because of this many others have evolved a range of methods of defence. Although most fish leave their eggs and young to look after themselves, some species protect their young.

Swim bladder

Bony fish have a swim bladder containing air, and they are able to fill and empty it at will. In many bony fish the swim bladder controls buoyancy, allowing the fish to move up and down in the water.



Inside a fish

Flatfish

Flatfish spend most of their lives lying on their sides, half buried and camouflaged in the sand on the seabed. Like most fish, the young develop in eggs. They hatch into normal larval fish that swim "the right way up" in the plankton.

10 days old



1 The larval fish has an eye on either side of its head.

17 days old



2 One eye gradually "migrates" to the other side of the head.

35 days old

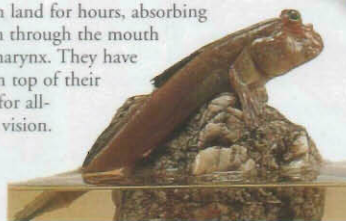


3 An adult flatfish lies on one side. Its eyes are on top.

Where fish live

There are fish living wherever there is water. Some fish live in oceans, the largest numbers living in the shallow seas of the continental shelf. The most brightly coloured fish live on coral reefs. Other fish live on muddy, sandy, or rocky seashores, in estuaries, in rivers and streams, and even in temporary puddles.

Mudskipper



On land

Inhabitants of Indo-Pacific mudflats and mangrove swamps, mudskippers are able to leave the water. They can stay on land for hours, absorbing oxygen through the mouth and pharynx. They have eyes on top of their heads for all-round vision.

Fins

Most fish have a dorsal fin, paired pectoral and pelvic fins, and a tail for movement. In some fish, fins have become specialized as lifting foils, walking legs, suckers for holding on, or poisoned spines for protection.

Scales

Most fish have a covering of backward-facing scales that help to streamline them. Bony fish have either flat, oval, or square overlapping scales, while sharks have tooth-like structures buried in the skin.



European carp

Fish features

Fish have a number of features in common. They breathe through gills, and are generally streamlined in shape with paired body muscles along each side. They have a tail for propulsion, fins for steering, and scales for protection. Their heads contain paired eyes and an obvious mouth with teeth.

Gills

Almost all fish have gills for breathing. The sharks and rays have paired gills in the throat, with openings to the outside known as gill slits. Bony fish have paired gills at the back of the head, with one opening covered by a flap of skin known as the operculum.



Fish groups

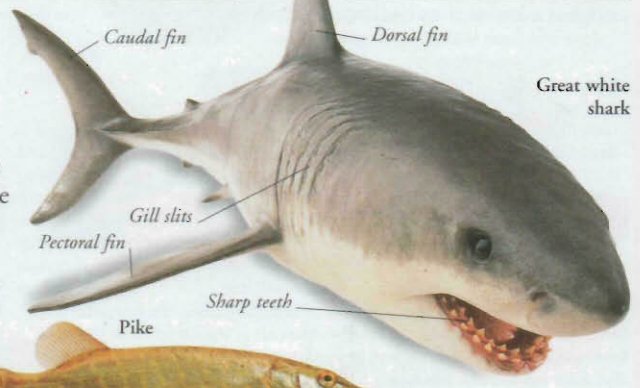
The fish are divided into three groups: jawless and primitive fish (cyclostomes), which include the lampreys and hagfish; cartilaginous fish (elasmobranchs), which include the sharks, rays, and ratfish; bony fish (teleosts), which include the more familiar fish, such as herring, cod, plaice, trout, eels, goldfish, sticklebacks, and guppies.

Jawless fish

Hagfish and lampreys have funnel-shaped mouths. Lampreys attach themselves to other fish with their mouths and rasp away flesh with their teeth. Hagfish are scavengers.



Sea lamprey



Great white shark

Pike



Bony fish

These fish are divided into two groups – those with jointed bony fin rays, such as most fresh- and saltwater fish, and those with fleshy fin lobes, such as lungfish.

Cartilaginous fish

Sharks and rays have large mouths with many teeth in rows that are continually being replaced. Their skeletons are made of cartilage (gristle), instead of bones.

Red mullet



Rainbow trout



Oceans

The oceans provide a range of habitats for fish. Light-producing fish live in the deepest ocean trenches, while other fish live near hot volcanic vents. Some, such as sharks, roam the open oceans searching for smaller fish to eat.

Fresh water

Fish live in fast-flowing streams, slow-moving rivers, ponds, and lakes, and are suited to their habitat. Some have to be powerful swimmers or have suckers for holding on to stones. Others live in shoals or are camouflaged to avoid predators.